

Urban Influence on Network Centrality Among Al-Qaeda Operatives

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1. Introduction

Urban Influence on Network Centrality Among Al-Qaeda Operatives. On September 11th, 2001, the terrorist organization Al-Qaeda carried out their large scale attack plan, directing 4 hijacked planes from American Airlines and United Airlines to crash into many politically vital buildings in America. As the hijackers' plan was to hit Manhattans' World Trade Center (WTC) North Tower and South Tower, the Pentagon, and the Capitol, they succeeded with almost all targets, except the Capitol. Among the consequences of the 4 crashed planes, the 2 targeting the Twin Towers of WTC exerted the most detrimental result with approximately 2,750 deaths (Holmes, 2005). Since then, the event known as 9/11 has remained a devastating incident in American history, noting Al-Qaeda as one of the most notorious terrorist groups to be known of (The Editors of Encyclopaedia Britannica, 2024).

Being the culprit behind such a significant event, Al-Qaeda became a famously discussed and studied alliance, resulting in numerous papers published regarding their network. Hence, not only were the 19 terrorists who directly hijacked the planes identified, but also 18 additional individuals who were involved in support functions such as logistics and finance, completing a completed network of 37 terrorists who were believed to relate to the 9/11 attacks (Krebs, 2001). Intrigued by the way the network connects them altogether to be capable of carrying out such a massive attack, we decided to look at the underlying reasons that might predict the centrality of each terrorist.

2. Problem Statement

As Matz and Harari (2021) claimed that people's personalities and choices are largely influenced by the environment surrounding them, we believe that in Al-Qaeda's context, individuals who were born in minor regions are more likely to be influenced by criminals, and therefore more likely to be a terrorist. Our reasoning behind this is that people born in smaller cities tend to have less privileges than those born in big ones (Squires & Kubrin, 2004). According to Curto et al. (2023), less fortunate conditions lead to individuals having higher chances to commit crimes, thus increasing their likelihood to become terrorists in this situation.

2.1 Research Question

The paper will investigate the following research question: Does the sizes of the cities where the terrorists are born predict their centrality in the network of 37 Al-Qaeda terrorists involved in the 9/11 event? We expect to find out that the smaller the cities, the more central the terrorists are in the network.

3. Literature Review

There have been various papers published regarding the topic of terrorism, with the purposes of not only to predict and prevent future potential attacks, but also to understand terrorists better for research. One of the topics touched upon was regarding the factors that might largely impact the chances of individuals engaging in terrorism crimes, suggested by Ljubic et al. (2017). In particular, the authors looked into the criminals' socio-economics status, which is measured using education level and unemployment state in their background. The findings met their expectation, as they found a remarkable number of criminals, including terrorists who not only were school dropouts, which indicates low education level, but also were unemployed at the moment of committing crime. Additionally, they also investigated the criminals' criminal record, and revealed a significant percentage of terrorists had committed violent crimes in the past. The study showed education level, employment status, and criminal past as an element affecting terrorists' choices.

Looking more into the network of 9/11 attacks specifically, Krebs (2001) conducted a research into the 37 terrorists involved. With the purpose to understand further their personal connections which plays an important role in managing the event, he depicted a completed and detailed network based on many different settings which are most likely to be where the terrorists knew each other from in advance. Namely, he noted all ties between the ones who study at the same class, same university to be the strong relationship in the network. He also put the relationships of terrorists who were identified travelling together into the list of links. Despite being rated as weak ties, he also looked at individuals who conducted financial transactions to and from each other. Having thought of a variety of ways how the terrorists can possibly know each other, he suggested potential benefits from future necessary law enforcement which can help identify attacks from the planning phase, preventing them from happening. The comprehensive network has also provided us with the required information to analyse the research question.

Similar to Krebs, Walter et al. (2023) also dived into the topic of influences on the network. However, he studied the spatial context in-depth by demonstrating models of all recorded travel geography of the 9/11 terrorists. He developed this by using the concepts of co-location, which directs to terrorists who stayed at the same cities at the same time, and co-destination, meaning the terrorists that share the same flight destination, though not necessarily the same time. Nevertheless, results show that 9/11 attacks did not require the terrorists to

physically meet to carry out the plan or discuss, which proves that identifying terrorism patterns appeared to be more challenging.

To get to know the target of the research, Bakker and Boer (2007) established a paper examining thoroughly Al-Qaeda as a terrorist group. It was discovered that they originated from an organization formed during the Afghan-Soviet War to fight for Afghanistan. Their ideology is to root for the establishment of a global Islamic state, and eliminate all foreign impact on Muslim countries. As their ideology actively advocates for violent jihad to combat the aforementioned impact, Al-Qaeda has become a global group attracting jihadists from worldwide. Their idea easily appeals to many Muslims because when they depict themselves as the defenders of the true Islam, the Muslims who feel alienated by the political situations are more likely to align themselves with their ways.

Regarding the motivation why Al-Qaeda specifically carried out 9/11 attacks on the United States, it has been analyzed by Byman (2015). With the aforementioned belief in themselves as the defenders of Islam against Western influence, their leader then, Osama bin Laden believed that by attacking the United State would drive them to retrieve their military from the Muslim community. Moreover, having been frustrated with United States support for Israel in the war with Palestine, they also perceive the attack as a punishment for their actions.

4. Methodology

This study aims to investigate the relationship between the geographical location of Al-Qaeda operatives and their centrality within the network. R Studio is used to process network connections and the terrorists' addresses data, collected through a comprehensive investigation of the demographical databases of different economies, to aid us in our network analysis.

4.1 Descriptive statistics

This study aims to investigate the relationship between the geographical location of Al-Qaeda operatives and their centrality within the network. R Studio is used to process network connections and the terrorists' address data, collected through a comprehensive analysis of the demographical databases of different economies, for analysis. To quantify various facets of an operative's influence within the network, the dependent variable, centrality, is measured using eigenvalue centrality, betweenness centrality, and degree centrality. Eigencentrality identifies nodes connected to other well-connected nodes, indicating overall influence (Newman, 2010). Betweenness centrality measures the frequency a node appears on the shortest paths between

other nodes, reflecting control over information flow (Adday et al., 2024). Degree centrality counts direct connections, highlighting immediate network activity (Newman, 2010). Independent variables include suburban level, crime rates, nationality, and income level. Data preparation involves gathering the initial home addresses of the terrorists, integrating crime statistics, and calculating centrality measures. Descriptive statistics and visualisations (network graphs) will help us identify patterns. Multivariate regressions and correlation analyses test the hypothesis that according to the size of the city, operatives exhibit higher centrality. T-tests will compare mean centrality scores across different categories. Robustness checks and validation will allow us to ensure reliability, addressing potential limitations like data quality and generalizability.

The Data Sources for the particular research have been derived from two primary sources. The first one is Terrorist addresses, a data set that includes detailed information on the residential addresses of identified Al-Qaeda operatives. The data is obtained from various intelligence reports, captured documents, and public records. The addresses are geocoded to enable accurate classification into urban or suburban categories so as to theorise our research question. Each terrorist's location has been rated on a scale of 1-5 to quantify the suburban level, with 1 being highly urban and 5 being highly suburban. Moreover, Data on their nationality, the population density of the cities in which they grew up, income levels, crime rates and GDP per capita of these cities have been gathered to help support our economic analysis. Furthermore, another dataset that's been used is the Terrorist network Data which maps the relationships and interactions among Al-Qaeda operatives, providing a basis for network analysis. The network data is compiled from communication logs, social media connections, and recorded interactions, depicting the intricate web of connections within the terrorist organisation which allowed us to establish network connections between the involved personnel (Krebs, 2001).

Centrality within the network, which evaluates the significance of Al-Qaeda agents using a variety of measures, is the dependent variable in this study. One of the metrics that is employed is eigenvalue centrality, which takes into account the quantity and quality of a node's connections (Newman, 2010). This allows for the identification of operatives who are linked to other highly influential nodes, hence demonstrating their overall influence within the network. The frequency with which an operative occurs on the shortest paths between other nodes is measured by another metric called betweenness centrality, which suggests substantial control over information flow (Newman, 2010). Conversely, degree centrality measures how

many direct connections an agent possesses, giving information about their recent network interactions and activities (Newman, 2010). Together, these centrality metrics allow for the evaluation of several facets of an agent's notoriety and sway inside the Al-Qaeda network.

The study's independent variables include a number of significant elements that could affect where and how central Al-Qaeda agents are located within their network. Suburban level serves to distinguish between living situations that might affect an operative's network access and mobility. It is classified as either urban or suburban depending on population density and urban planning data. As a continuous variable, crime rates offer information about the safety and security of an operative's home, which may have an impact on their network strategy decisions. Birthplace provides context about an operative's past and possible connections to particular Al-Qaeda locations or networks. It is most likely meant to capture nationality or origin. The last continuous variable is income level, which indicates an operative's financial means and might impact their living conditions and accessibility to particular locations, thus, affecting their centrality and network interactions. When taken as a whole, these independent variables seek to offer a thorough framework for comprehending the socioeconomic and geographic factors influencing the prominence of Al-Qaeda agents inside their network.

When researching Al-Qaeda operators, comprehensive data analysis begins by summarizing essential statistics such as mean, median, standard deviation, and range for continuous variables such as income and crime rates. The fundamental understanding of the data's basic patterns and variability is given by these statistics. At the same time, frequency distributions are used to analyze categorical variables such as nationality and suburban dwelling to determine how operatives are distributed across various groups and look for any noteworthy trends or anomalies. Furthermore, Network graphs serve a crucial role in helping to visualize the complex structure of the terrorist network. Based on measures like betweenness centrality and eigencentrality, these graphs emphasize important actors and give a clear picture of the dynamics of the network (Cowan, 2023). This study intends to contribute to a deeper scholarly knowledge of terrorist organizational dynamics by utilising these analytical and visualisation approaches to reveal underlying socio-economic and geographic elements that are impacting the importance of Al-Qaeda operators within their network.

The primary goal of this study is to examine the hypothesis that city size significantly influences centrality within the Al-Qaeda network. Several statistical techniques will be used

to investigate this relationship in detail. First, the strength and direction of linear correlations between continuous variables, like income level and centrality scores, will be evaluated through correlation analysis using the Pearson Correlation Coefficient. The purpose of this investigation is to find statistically significant correlations that may show the relationship between network prominence and socioeconomic characteristics.

Moreover, the study will employ multiple linear regression analysis to investigate the combined influence of several independent factors, such as Population density of the operatives birthplace, GDP per capita of their birth place and crime rates, on centrality. This method offers a thorough grasp of how many elements jointly shape operatives' centrality within the network by taking into account potential confounding variables. Developing theories is essential to this approach. The alternative hypothesis (H_1) contends that there is a substantial relationship between city size and centrality, contrary to the null hypothesis (H_0), which holds that there is no significant association at all. T-tests will complement these analyses by comparing mean centrality scores between urban and suburban operatives, determining if significant differences exist based on geographic location.

Analyzing the statistical significance of the data—usually denoted by p-values less than 0.05—will help determine whether to reject the null hypothesis and favor the alternative in the interpretation of the results. The quantification of effect sizes can provide light on the practical significance of findings by measuring the magnitude of relationships that have been identified. Furthermore, assessing model fit using metrics like R^2 value will determine the extent to which regression models account for fluctuations in centrality, providing information about the resilience and explanatory ability of the models.

The methodology described herein provides a thorough framework for assessing the relationship between Al-Qaeda operatives' geographical location and their centrality in the network. This project intends to provide useful insights on the strategic deployment of operators and improve predictive capacities for future counter-terrorism efforts by combining data from multiple sources, employing rigorous statistical methods, and adhering to ethical standards. The results will assist influence policy decisions and provide strategies for disrupting terrorist networks.

4.2 Analysis Plan

Analysis Plan Template	
Date 26th May 2024	Name of the project: City size as a Predictor of Centrality Among Al-Qaeda Operatives
Team Members: Thanthirige Navodha Costa Giang Nguyen	
Introduction to the topic and Research Question	
Brief introduction: Following 9/11, al-Qaeda gained global notoriety. Key terrorists may relocate to safer suburbs, challenging the belief they prefer urban cores for anonymity. Investigating this could help predict their future locations.	
Research Question: Does the sizes of the cities where the terrorists are born predict their centrality in the network of 37 Al-Qaeda terrorists involved in the 9/11 event?	
Hypothesis: The size of the city of Al-Qaeda terrorists predicts their centrality in the network; those in smaller cities are likely to have higher centrality scores.	
Data details	
Data set(s) used: Terrorist addresses and terrorist network	Analysis Software: R studio
Dependent variable(s): (what you are trying to explain) Centrality	
Independent Variables: (what you think would explain it) CitySize Crime rates Population density GDP per capita Income level	
Proposed Analytical Strategy	
Data Manipulation: Do you need to manipulate your data? For example, you would like to use per capita income as variable, but you don't have it in the data set. However, you do have variables with household income and household size, so you can use those two to generate your variable of interest. Yes. We will be manipulating the data of suburb level according to education level and income level.	
Data Visualization: if you draw a scatterplot or a boxplot, can you spot a pattern? Yes, we do. We can potentially spot detailed patterns in the data indicating the characteristics and strategic choices of central figures in the network by using network graphs.	
Data Analysis: Will you use multivariate regressions, correlations, ANOVA, ANCOVA, etc.? What graphs or tables will you include? we will be using: multivariate regressions and correlation. We will be including graphs of the network and statistical tables to help support our hypothesis.	

5. Findings

The study sought to evaluate the relationship between geographical parameters and the centrality of Al-Qaeda operatives in their network. This research employed descriptive statistics, a correlation matrix, and a multivariate regression analysis, with centrality as the dependent variable and city size, crime rate, population density, and GDP per capita as the independent factors. We also create a network graph of the Al Qaeda militants who participated in the 9/11 attacks depending on their city size (Krebs, 2001).

Descriptive statistics offered a basic comprehension of the dataset. The mean, median, and standard deviation of the continuous variable city size was calculated. The mean centrality score throughout the sample was fairly balanced, with a standard deviation showing substantial fluctuation in operative centrality within the network. Notably, operators in larger cities had higher average centrality scores than those in smaller cities, indicating that city size may influence network importance. As shown in *Figure 1* below, The mean of 2.07 gives us a sense of the central point of the data, while The standard deviation of 1.288 tells us that there is a moderate spread in the data around this mean. Most values are within 1.288 units of the mean, which implies a reasonable but not excessive level of variability in the data, indicating that most cities where the terrorists resided were on a larger scale.

```
[1] "Mean: 2.7027027027027"
> sd_city_size <- sd(city_size_level)
> print(paste("Standard Deviation:",sd_city_size))
[1] "Standard Deviation: 1.28808352181027"
```

Figure 1: Mean and Standard deviation based on City size

After running the network statistics in R, *Figure 2* below reveals the relevant figures conjured from the terrorists' address dataset which revealed some crucial insights into the network structure of Al-Qaeda operatives. For example, the network's average degree is 4.59, which means that each node (operative) is directly connected to around 4.6 other nodes. This indicates a moderately linked network. The average path length of 2.61 is the average number of steps needed to connect any two nodes, demonstrating rather high connectedness Cowan,R (2023). The network density of 0.1276 implies that just 12.76% of all possible connections

are present, indicating a sparse network. An average clustering coefficient of 0.5231 shows that the network has a moderate level of clustering, with 52.31% of a node's neighbors also being neighbors. The average betweenness centrality of 29 indicates substantial control over information flow to critical nodes in the network suggesting the terrorist operatives were able to stay in contact, regardless of their geographical location.

	Name	Value
AverageDegree	AverageDegree	4.5945946
AveragePathLength	AveragePathLength	2.6111111
Density	Density	0.1276276
AverageClustering	AverageClustering	0.5230555
AverageBetween	AverageBetween	29.0000000

Figure 2 : Network statistics of the terrorist network

According to *Figure 3*, The correlation matrix revealed a weak negative association between city size and centrality (Pearson correlation coefficient: -0.0221). This suggests that as city size increases, centrality diminishes slightly, however the relationship is not strong. This conclusion implies that operatives in larger cities may not necessarily have more central positions within the network. This small link suggests that other factors, besides city size, may have a greater impact on centrality.

	Small_city	Centrality
Small_city	1.00000000	-0.02208182
Centrality	-0.02208182	1.00000000

Figure 3: Correlation matrix

The regression model used in this study is as follows: $Centrality = \beta_0 + \beta_1 CitySize + \beta_2 + \beta_3 + \beta_4 + \epsilon$, where β_0 is the intercept, $\beta_1, \beta_2, \beta_3, \beta_4$ are the coefficients for each predictor, and ϵ is the error term. The regression analysis as shown in *Figure 4* below revealed several key insights. The intercept was significant ($p < 0.05$), indicating that when all other predictors are zero, the centrality measure is approximately 5.690. This suggests a baseline centrality level inherent to the operatives irrespective of other factors.

However, none of the independent variables—city size, crime rate, population density, and GDP per capita—had statistically significant correlations with centrality (p -values > 0.05). The coefficients for City size ($p = 0.7022$), Crime rate ($p = 0.5573$), Population density ($p = 0.9801$), and GDP per capita ($p = 0.9405$) were as follows. These findings indicate that there is no evidence in this dataset that geographic and economic factors have a substantial influence on the centrality of Al-Qaeda operatives.

The residual statistics illustrate the model's weaknesses, with a wide range of centrality values. The residual standard error was 3.539 on 32 degrees of freedom, demonstrating significant variability in centrality scores that the model did not account for. The low multiple R-squared value (0.02482) and negative adjusted R-squared (-0.09707) indicate that the model does not account for much of the variability in centrality. The high p -value (0.9345) of the F-statistic indicates that the entire model is not statistically significant.

These findings suggest that, based on existing data, geographic and economic characteristics such as city size, crime rates, population density, and GDP per capita have no influence on the centrality of Al-Qaeda operatives inside their network. This could imply that other unexplored aspects, such as organizational dynamics, personal characteristics, or operational duties, are more important in defining an operative's centrality. For example, an operative's access to resources, training, or personal ties within the organization may be more important than their geographic or economic background (Bettencourt et al., 2007).

Furthermore, the lack of significance in the findings may indicate that Al-Qaeda operatives are strategically positioned based on operational effectiveness and loyalty rather than physical location. These operatives could be assigned to tasks where their talents and dependability are more important than the socioeconomic circumstances of their upbringing. This could represent the organization's agility and strategic priorities that put operational efficiency ahead of other concerns (Sageman, 2004).

```

Call:
lm(formula = Centrality ~ CitySize + CrimeRate + PopulationDensity +
    GDPperCapita, data = regre_data)

Residuals:
    Min       1Q   Median       3Q      Max
-4.1860 -1.9658 -1.0820  0.5484 10.0852

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   5.690e+00  2.232e+00   2.550   0.0158 *
CitySize      -2.135e-01  5.533e-01  -0.386   0.7022
CrimeRate     -1.949e-01  3.286e-01  -0.593   0.5573
PopulationDensity -4.169e-06  1.659e-04  -0.025   0.9801
GDPperCapita  -2.364e-06  3.140e-05  -0.075   0.9405
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.539 on 32 degrees of freedom
Multiple R-squared:  0.02482,    Adjusted R-squared:  -0.09707
F-statistic: 0.2036 on 4 and 32 DF,  p-value: 0.9345

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Figure 4: Multivariate regression analysis

The network plot as shown in *Figure 5* below of Al-Qaeda operators shows a clear hierarchical structure, with Mohamed Atta and Ramzi Binalshibh as major members, demonstrating their important leadership and coordinating duties. Their centrality enables them to manage information flow and direct operations. Tight clusters around these center figures indicate operational subgroups collaborating on specific tasks or regions, implying a defined division of responsibility within the network. Peripheral nodes, such as Wail Alshehri and Osama Awadallah, have fewer connections but may perform specific functions or serve as

intermediates between clusters. These nodes, in spite of being less central, are critical to the general operation of the network. Targeting central persons can severely impair network functioning, producing chaos and reducing coordination. Furthermore, knowing the dynamics among subgroups might aid in the identification of key operatives for targeted counter-terrorism measures, hence improving strategies to undermine Al-Qaeda's operational efficacy. This research offers useful insights for disturbing the network's cohesion and effectiveness.

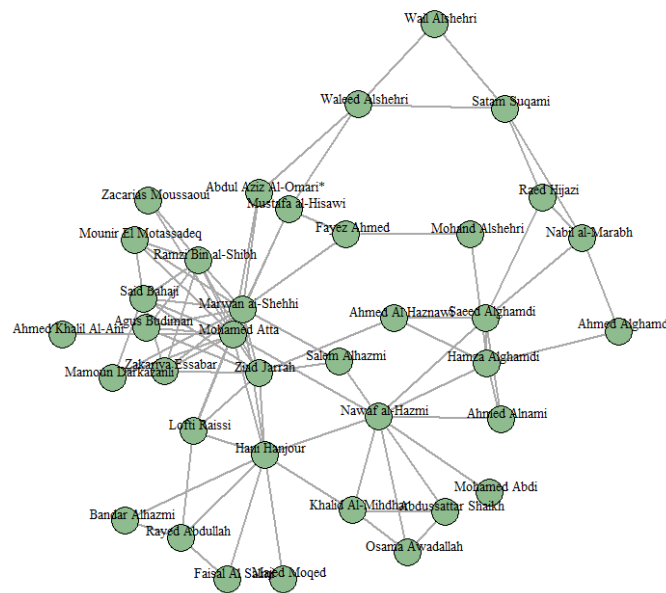


Figure 5: Network graph of the Al Qaeda Terrorists involved in the 9/11 attacks

6. Conclusion

To capture more complex correlations, future research should examine new variables as well as different methodological approaches, such as nonlinear models or machine learning. For example, incorporating characteristics such as personal background, training, length of involvement in terrorist operations, and connections to prominent persons may provide further information. Improving data quality by incorporating more comprehensive and recent datasets, as well as increasing sample size, may improve the conclusions' robustness and generalizability.

Furthermore, combining qualitative and quantitative analysis might yield more in-depth insights. Interviews with specialists, case studies of renowned operatives, and comprehensive network analysis employing modern technologies could all disclose nuances that quantitative methods alone cannot. Such mixed-method approaches would help to validate findings and offer a comprehensive grasp of the factors that influence centrality in terrorist networks.

To conclude, the findings of this study add to our understanding of Al-Qaeda operatives' strategic deployment and highlight the importance of a diverse strategy to counter-terrorism activities as well indicating that city size is not a good measure of centrality in a network of terrorists.. This research study intends to influence policy decisions and strategies for disrupting terrorist networks by merging data from many sources and using rigorous statistical methodologies. Although the current findings indicate the need for additional in-depth analysis to properly comprehend the dynamics at work, they also highlight topics for further research. Future studies should strive to improve on these findings by using more diverse data sources and sophisticated analytical approaches to better understand and disrupt terrorist organizations' operational plans, such as Al-Qaeda.

7. References

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